Statistical chronometry of meteorites revisited

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Abstract

The chronology of the early solar system rests on extinct (^{26}Al–^{26}Mg, ^{53}Mn–^{53}Cr, ^{182}Hf–^{182}W) and extant (^{207}Pb–^{206}Pb) chronometers. These should agree with each other if dating the same events, such as isotopic closure of these systems in a rapidly cooled achondrite. If multiple achondrites formed on different parent bodies, the chronometers should be concordant across them, assuming isotopically homogeneous values of ^{26}Al/^{27}Al, ^{53}Mn/^{55}Mn and ^{182}Hf/^{180}Hf. Testing the homogeneity of these short-lived radionuclides (SLRs) is often done by comparing a few samples, but really requires a statistical framework considering many meteorites simultaneously. Here we discuss two similar statistical approaches. The first is that of Desch et al. (2023a,b: doi:10.1016/j.icarus.2023.115607 and doi:10.1016/j.icarus.2023.115611), which calculates the time of formation of an achondrite after the start of the solar system ($t = 0$, which happened a time t_{SS} ago), in different isotopic systems, and takes their weighted mean as the best estimate of the formation time. This model was applied to 38 dates across 14 multiply-dated achondrites, adjusting Pb–Pb ages for ^{238}U/^{235}U fractionation during mineral formation. We correct an error in their calculation of the goodness-of-fit parameter and demonstrate that the adjusted formation times show a small but statistically significant degree of overdispersion with respect to their analytical uncertainties. We also introduce a second approach to statistical chronometry, which is to calculate times of formation before the present, essentially converting dates into 'Pb–Pb times', using a single offset parameter per isotopic system, estimated by a weighted regression with a fixed unit slope. Applying the second algorithm to the same data compilation yields the same conclusion of statistically significant overdispersion. Extending the regression to a random-effects model allows the excess scatter to be quantified. The overdispersion of the Pb–Pb ages relative to the extinct chronometers is greatly reduced after the corrections of Desch et al. (2023) for uranium isotopic fractionation, from 450–900 kyr to 50–380 kyr. Whether this remaining overdispersion is astrophysical or analytical in origin is unresolved. One explanation for this remaining overdispersion might be variability of SLR abundances. If all variations are attributed to this, it would require 0.5 to 7.5% (or 40 to 130%) variations in the ^{26}Al/^{27}Al ratio, or 2 to 7.5% (or 5 to 22%) variations in the ^{53}Mn/^{55}Mn ratio, or 1.4 to 6.8% (or 3 to 10%) variations in the ^{182}Hf/^{180}Hf ratio (values in parentheses referring to the uncorrected Pb–Pb ages). These relatively small variations are difficult to rule out on purely astrophysical grounds, but would not explain the persistent overdispersion even among angrites, which derive from the same parent body. This suggests that systematic variations in analysis are at least partly responsible for the overdispersion of achondrite formation times. Routine measurement of ^{238}U/^{235}U in the mineral phases used for Pb–Pb dating, and improvements in development of isochrons, would test this interpretation.

1 Introduction

The chronology of the early solar system can be constrained from the radiogenic decay products of extinct (^{26}Al, ^{53}Mn, ^{182}Hf) and extant (^{235}U, ^{238}U) nuclides in meteorites. The extinct nuclides form the basis of relative chronometers, which can be used to estimate the age difference between pairs of meteorites. In

principle, the extant nuclides can be used to anchor meteorites in absolute time, counting back from the present. However, in practice the precision of such absolute dates is limited by the analytical uncertainty of the decay constants (Ludwig, 2000). For the $^{207}Pb/^{206}Pb$ method, this uncertainty amounts to ca. 9 Myr ($2\,\sigma$) at 4567 Ma (Figure 1). The decay constant uncertainty is greatly reduced when calculating the difference between two similar $^{207}Pb/^{206}Pb$ dates (Figure 1). If a different value of the decay constant is assumed, the ages of both samples are increased or decreased by a similar amount. The age difference is therefore much more certain than ±9 Myr. For example, a 10 Myr age difference near the onset of the solar system is associated with a decay constant uncertainty of only 7 kyr, as long as all the samples are dated assuming the same decay constants. In that case, when comparing $^{207}Pb/^{206}Pb$ dates within the 4550–4568 Ma age range, decay constant uncertainties can be safely ignored, turning the $^{207}Pb/^{206}Pb$-method into a relative chronometer with similar precision as the extinct chronometers.

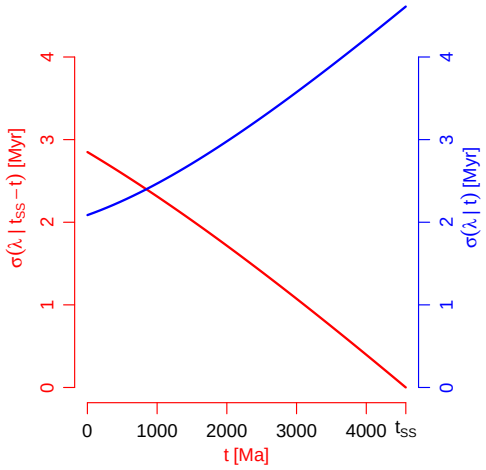


Figure 1: (blue) uncertainty in the inferred Pb–Pb age of a sample with true age t, due to the uncertainty in the decay constants of ^{238}U and especially ^{235}U; (red) uncertainty in the difference between the inferred Pb–Pb age of a sample with true age t, and a second sample with age t_{SS}, Pb–Pb dated using identical decay constants which are uncertain. When used as a relative chronometer to date differences in formation times between two objects, the 1σ uncertainty is small (roughly 1% of the age difference); but when used as an absolute chronometer, the $2\,\sigma$ uncertainty is about 9 Myr.

For a rapidly ($\ll 10^5$ yr) cooled achondrite in which all isotopic systems reached closure simultaneously, the ages or formation times derived from each system should agree, assuming the Pb–Pb age of the solar system, t_{SS}, and a particular set of initial values of $^{26}Al/^{27}Al$, $^{53}Mn/^{55}Mn$, and $^{182}Hf/^{180}Hf$. The ages or formation times of a second rapidly cooled achondrite also should agree with each other. If the short-lived radionuclides were uniformly distributed in the solar nebula, the ages of both achondrites would be self-concordant and concordant with each other, for the same set of assumed parameters. The best tests of homogeneity extend this idea to a basket of achondrites date across multiple systems, comparing multiple samples simultaneously, using a 'statisical chronometry' approach.

This paper presents two approaches to this problem. A first approach is to establish a chronology for the early solar system counting forward in time from the start of the solar system, a time t_{SS} ago, using the scatter of double-, triple- and quadruple-dated meteorites around their weighted means (Section 2.1). A second approach is to build the statistical chronology counting backward in time, starting from the present and using linear regression instead of weighted means (Section 2.2).

In Section 3, we apply both approaches to a dataset of Al–Mg, Mn–Cr, Hf–W and revised Pb–Pb ages across 14 achondrites, compiled by Desch et al. (2023a,b). Previous statistical analysis of this database using the first approach suggested concordance of the Al–Mg, Mn–Cr, Hf–W and Pb–Pb systems. After fixing an error in the calculations of Desch et al. (2023a,b), we find that in fact the data show a small but statistically significant amount of excess dispersion. In Section 4, we quantify this overdispersion by modifying the linear regression method, and discuss the possible causes of the overdispersion.

2 Methods

Before discussing the methods, we define the following terms.

$t_{\rm SS}$: the age the solar system, measured backward from the present;

t: the time elapsed since the start of the solar system, measured forward in time;

$t_{\rm Al}$, $t_{\rm Pb}$, $t_{\rm Mn}$, $t_{\rm Hf}$: the Al–Mg, Pb–Pb, Mn–Cr and Hf–Cr ages, respectively, measured backwards in time from the present;

$\Delta t_{\rm X} = t_{\rm SS} - t_{\rm X}$, where $t_{\rm X} \in \{t_{\rm Al}, t_{\rm Pb}, t_{\rm Mn}, t_{\rm Hf}\}$;

λ_{26}, λ_{53}: the decay constants of ^{26}Al and ^{53}Mn, respectively;

$\left[{}^a{\rm X}/{}^b{\rm X}\right]_{\rm SS}$: the ${}^a{\rm X}/{}^b{\rm X}$ (e.g., $^{26}{\rm Al}/^{27}{\rm Al}$)-ratio of the solar system at $t = 0$, a time $t_{\rm SS}$ ago;

$\left[{}^a{\rm X}/{}^b{\rm X}\right]_0$: the ${}^a{\rm X}/{}^b{\rm X}$-ratio of a meteorite at the time of its formation;

$\sigma[*]$: the analytical uncertainty (estimated standard deviation) of $*$;

χ^2: the chi-square statistic for homogeneity;

ν: the number of degrees of freedom of the chi-square distribution under the null hypothesis;

MSWD: the mean square weighted deviation, defined as MSWD $= \chi^2/\nu$.

2.1 First Approach: $t_{\rm SS} \rightarrow t$

Desch et al. (2023a) define $t_{\rm SS}$ as the Pb–Pb age that would be inferred for a sample that isotopically closed at time $t = 0$, defined to be when the $^{26}{\rm Al}/^{27}{\rm Al}$ in the solar system was $(^{26}{\rm Al}/^{27}{\rm Al})_{\rm SS} \equiv 5.23 \times 10^{-5}$, the value widely accepted as recorded by CAIs (**?**). Because the Pb–Pb system could be reset by transient heating events (without resetting the Al–Mg system), Desch et al. (2023a) did not consider this age to be recorded by any CAI. Therefore they regarded this as a free parameter that must be determined, by finding the value of $t_{\rm SS}$ that minimizes the differences between $\Delta t_{\rm Al} = \frac{1}{\lambda_{26}}\left(\ln\left[\frac{^{26}{\rm Al}}{^{27}{\rm Al}}\right]_{\rm SS} - \ln\left[\frac{^{26}{\rm Al}}{^{27}{\rm Al}}\right]_0\right)$ and $\Delta t_{\rm Pb} = t_{\rm SS} - t_{\rm Pb}$, for each achondrite. To be more specific, given a sample of A achondrites with both Al–Mg and Pb–Pb dates, Desch et al. (2023a) found the value that minimizes the chi-squared statistic

$$\chi^2 = \sum_{i=1}^{A} \left(\frac{\Delta t_{{\rm Al},i} - \Delta t_i}{\sigma_{{\rm Al},i}}\right)^2 + \left(\frac{\Delta t_{{\rm Pb},i} - \Delta t_i}{\sigma_{{\rm Pb},i}}\right)^2, \tag{1}$$

where Δt_i is the error-weighted mean of $\Delta t_{{\rm Al},i}$ and $\Delta_{{\rm Pb},i}$. By minimizing χ^2, the optimal value of the free parameter $t_{\rm SS}$ (which shows up in each $\Delta t_{{\rm Pb},i}$) is found.

If the A achondrites are all concordant, and the stated analytical uncertainties are the only source of dispersion, then χ^2 should follow a chi-square distribution with ν degrees of freedom: low values would be consistent with isotopic homogeneity, but high values $\chi^2 \gg \nu$ would indicate overdispersion of some cause. The number of degrees of freedom is $A - 1$. Essentially, there is one comparison to be made for each achondrite, and there is one free parameter ($t_{\rm SS}$). Desch et al. (2023a) incorrectly used $\nu = 2A - 1$, assuming that the deviations of $\Delta t_{{\rm Al},i}$ and $\Delta t_{{\rm Pb},i}$ from Δt_i were statistically independent. In fact, by the definition of Δt_i, they are perfectly anti-correlated, and only one degree of freedom is introduced per achondrite, not two. Another way to look at it is that there are $2A$ variables being fit, with the 1 parameter $t_{\rm SS}$, but also A free parameters that are the weighted mean ages Δt_i, so that $\nu = 2A - A - 1 = A - 1$.

Equation 1 can be generalized to more than two chronometers, as was done by Desch et al. (2023b). In this case, it is not necessary that Al–Mg and Pb–Pb dates both exist for every achondrite; Desch et al. (2023b) included in their compilation 14 achondrites that had Pb–Pb ages and data for at least one other system (Al–Mg, Mn–Cr, or Hf–W). Here the fit must also simultaneously optimize $t_{\rm SS}$, $(^{53}{\rm Mn}/^{55}{\rm Mn})_{\rm SS}$, $(^{182}{\rm Hf}/^{180}{\rm Hf})_{\rm SS}$, and λ_{53}. The error-weighted mean Δt_i is based on whatever formation times are available

for each achondrite. When assessing the goodness-of-fit, Desch et al. (2023b) assumed $\nu = N - 4$, based on N formation times across the A achondrites, minus the 4 free parameters. However, the same logic about the weighted means as an introduced free parameter applies, and the correct number of degrees of freedom is $\nu = N - A - 4$.

2.2 Second Approach: $t \leftarrow$ present

It is worth considering whether a different approach might reduce the number of free parameters, and allow a different assessment of whether or not the formation times from Desch et al. (2023b) are overdispersed. The first approach defines $t_{\rm SS}$ as the Pb–Pb age inferred for a sample that isotopically closed at the time the ^{26}Al/^{27}Al ratio in the solar system (assuming isotopic homogeneity) was a fixed value $(^{26}\mathrm{Al}/^{27}\mathrm{Al})_{\rm SS} = 5.23 \times 10^{-5}$. As outlined above, this age is determined using a basket of multiply-dated achondrites to align the Pb-Pb and Al-Mg chronometers. Future updates to the basket would therefore result in new estimates for $t_{\rm SS}$.

A mathematically equivalent use of the basket would be to fix t_{SS} and treat $(^{26}\mathrm{Al}/^{27}\mathrm{Al})_{SS}$ as an unknown. Alternatively, the two quantities, t_{SS} and $(^{26}\mathrm{Al}/^{27}\mathrm{Al})_{SS}$, can be combined into a single free parameter by expressing the Al/Mg-dates in 'Pb/Pb-time', i.e. by measuring time from the present rather than from t_{SS}. Applying this method to each achondrite i:

$$t_{\mathrm{Al},i} = t_{\rm SS} - \frac{1}{\lambda_{26}} \ln\left(\frac{^{26}\mathrm{Al}}{^{27}\mathrm{Al}}\right)_{\rm SS} + \frac{1}{\lambda_{26}} \ln\left(\frac{^{26}\mathrm{Al}}{^{27}\mathrm{Al}}\right)_{0,i} = O_{26} + \frac{1}{\lambda_{26}} \ln\left(\frac{^{26}\mathrm{Al}}{^{27}\mathrm{Al}}\right)_{0,i}. \quad (2)$$

If the two isotopic systems are concordant, then we expect $t_{\mathrm{Al},i} = t_{\mathrm{Pb},i}$ for each achondrite, and for a plot of $t_{\mathrm{Al},i}$ vs. $t_{\mathrm{Pb},i}$ to be a line of slope 1, with unknown offset parameter, O_{26}, which is determined by linear regression. The resultant MSWD can be used to determine the goodness of fit. The number of degrees of freedom would be $\nu = A - 1$, as before. Any changes to the basket of meteorite dates would change O_{26}, and hence t_{Al}, but t_{Pb} would be unchanged. This approach can also be used to compare Pb–Pb ages to Mn–Cr and Hf–W formation times, just as they are compared to Al–Mg formation times.

3 Application to the Desch et al. (2023b) database

Desch et al. (2023b) compiled $N = 38$ Al–Mg, Mn–Cr, and Hf–W isochrons and Pb–Pb ages across $A = 14$ achondrites. In their detailed survey of the literature, they adjusted Pb–Pb ages as follows. First, they recalculated the Pb–Pb dates of some meteorites by retrospectively adjusting the ^{238}U/^{235}U-ratios by 0.3‰ to account for possible U-fractionation between pyroxene and whole rock, following Tissot et al. (2017). This adjustment was deemed necessary because ^{238}U/^{235}U measurements are almost always determined from whole-rock samples, whereas the Pb–Pb dating requires knowledge of the ^{238}U/^{235}U ratio in the mineral phase contributing the most radiogenic Pb, generally pyroxenes. Second, they recalculated ^{207}Pb/^{206}Pb vs. ^{204}Pb/^{206}Pb isochrons for a few achondrites for which certain aliquots were excluded from the regression without justification. They showed that equally valid alternative choices of which aliquots to include led to similar slopes (and ages), but with much greater uncertainty than had been reported. After this adjustment, Desch et al. (2023a,b) developed the database show in Table 1.

Table 1: Summary table for the database of Desch et al. (2023b). Headers use the following shorthand notation: $Al_0 \equiv \left(^{26}\mathrm{Al}/^{27}\mathrm{Al}\right)_0$, $Mn_0 \equiv \left(^{53}\mathrm{Mn}/^{55}\mathrm{Mn}\right)_0$, $Hf_0 \equiv \left(^{182}\mathrm{Hf}/^{180}\mathrm{Hf}\right)_0$; $t_{\rm orig}$ stands for the original Pb–Pb dates extracted from the literature; and $t_{\rm adj}$ represents the adjusted values used by Desch et al. (2023b).

sample	Al_0	$2\sigma(Al_0)$	Mn_0	$2\sigma(Mn_0)$	Hf_0	$2\sigma(Hf_0)$	t_{orig}	$2\sigma\,(t_{\mathrm{orig}})$	t_{adj}	$2\sigma\,(t_{\mathrm{adj}})$
D'Orbigny	3.93	0.39	3.23	0.03	7.15	0.17	4564.42	0.12	4563.24	0.21
SAH 99555	3.64	0.18	3.28	0.17	6.87	0.15	4564.614	0.131	4563.51	0.24
NWA 1670	5.92	0.59	2.85	0.92			4565.39	0.24	4564.02	0.66
NWA 1296					7.01	0.28	4564.2	0.45	4563.02	0.45
LEW 86010			1.35	0.05	4.8	0.42	4558.55	0.15		
NWA 4590			0.85	0.4	4.63	0.17	4557.76	0.38	4557.57	0.38
NWA 4801			0.96	0.04	4.52	0.16	4556.91	0.21	4556.63	0.28
Angra dos Reis			1.1	0.4	4.02	0.24	4556.45	0.29	4556.26	0.29
NWA 2999					5.43	0.34	4560.74	0.47	4560.55	0.47
Asuka 881394	13.07	0.56	3.86	0.23			4564.95	0.53	4564.76	0.53
Ibitira			1.06	0.5			4556.75	0.57	4556.56	0.57
NWA 7325	3.03	0.14					4563.9	1.7	4563.7	1.7
NWA 2976	4.05	0.15					4563.16	0.57	4563.16	0.57
NWA 6704	3.15	0.38	2.59	0.34			4562.76	0.26	4562.76	0.26

3.1 First approach

Applying the first approach to the $A = 7$ achondrites in Table 1, Desch et al. (2023a) found that a value of $tSS = 4568.42 \pm 0.24$ Myr minimizes Equation 1. They obtained an MSWD-value of $\chi^2/\nu = 0.98$ for the optimal fit, incorrectly using $\nu = 13$. In fact, $\nu = 6$, so that $\chi^2/\nu = 2.12$. It is worth noting that 95% of the time, for 6 degrees of freedom, $0.21 < \chi^2/\nu < 2.41$, so the 7 achondrites still can be considered concordant between the Al–Mg and Pb–Pb systems, even using the correct ν. Desch et al. (2023b) then optimized the fit using the full database, finding the values of t_{SS}, $(^{53}\mathrm{Mn}/^{55}\mathrm{Mn})_{\mathrm{SS}}$, $(^{182}\mathrm{Hf}/^{180}\mathrm{Hf})_{\mathrm{SS}}$ and the half-life of ^{53}Mn that minimize χ^2.

Applying an extended version of Equation 1 to the corrected data of Table 1, Desch et al. (2023b) found an optimal fit, for which they reported a χ^2-value of 47.94 for the 38 meteorite dates. This optimization led to essentially the same optimal value of $t_{\mathrm{SS}} = 4586.4 \pm 0.2$ Myr. Assuming $\nu = N - 4 = 34$ degrees of freedom, the p-value for the chi-squared test is 0.057, which is greater than 0.05 and fails to reject the null hypothesis that the ages are concordant. However, as discussed in Section 2.1, the actual number of degrees of freedom is $\nu = N - A - 4 = 19$. Using this correct value yields a p-value of 0.00043, resulting in a rejection of the null hypothesis.

Recognizing that the Hf–W date of NWA 4801 was quite divergent, and also derived from an achondrite with physical indications of disturbance to the Hf–W system, Desch et al. (2023b) present a second optimization without this point. Assuming $\nu = 37 - 4 = 33$, they reported $\chi^2 = 35.97$, resulting in a p-value of 0.33, which would be consistent with concordancy across the four isotopic systems and across the 14 achondrites. However, using the correct $\nu = N - A - 4 = 19$, the p-value is 0.011, which contradicts this conclusion.

3.2 Second approach

We now proceed to subject the data in Table 1 to the second approach. Figure 2 compares the original Pb–Pb dates with the Al–Mg, Mn–Cr and Hf–W dates, respectively. For each of the three pairs of measurements, the offset O_{26} (or O_{53} or O_{183}) was found by minimizing the weighted least squares misfit around the 1:1 line, using the algorithm of Vermeesch (2024). All three cases exhibit high MSWD-values and fail the chi-square test for homogeneity. This indicates that the data are overdispersed with respect to the analytical uncertainties.

There are several potential explanations for the excess dispersion of the data around the 1:1 line in Figure 2: (i) heterogeneity of the initial $(^{26}\mathrm{Al}/^{27}\mathrm{Al})_{\mathrm{SS}}$, $(^{53}\mathrm{Mn}/^{55}\mathrm{Mn})_{\mathrm{SS}}$, or $(^{182}\mathrm{Hf}/^{180}\mathrm{Hf})_{\mathrm{SS}}$ in the regions of the nebula sampled by each parent body. (e.g., **?**Krestianinov et al., 2023); (ii) failure of the different isotopic systems to close simultaneously, due to impacts or other events on the parent bodies that reset especially the Hf–W and

Pb–Pb systems; or (iii) a variety of analytical discrepancies leading to incorrect ages based on the data. The latter could include improper estimation of the isotopic fractionation of uranium between different phases, or inappropriate selection of aliquots to build the $^{207}Pb/^{206}Pb$ vs. $^{204}Pb/^{206}Pb$ isochrons, both of which would lead to dispersion in Pb–Pb ages. It could also include inappropriate outlier rejection in the Al–Mg, Mn–Cr or Hf–W isochrons, as well as the Pb–Pb isochrons (Desch et al., 2023a; Vermeesch, 2026). Finally, it could be due to inconsistent calibrations made during the collection of data to build those isochrons.

Desch et al. (2023a,b) adjusted the Pb–Pb dates in Table 1 in an attempt to correct for analytical errors (factor iii). This adjustment does indeed lower the MSWD-value considerably (Figure 3). Nevertheless, the three pairwise comparisons still fail the chi-square test. Therefore the conclusion from the second approach is still that the data are overdispersed, and for some reason the different chronometers are not concordant. In Section 4 we extend the second approach, to treat the various sources of heterogeneity as an unknown whose effect can be estimated from the data.

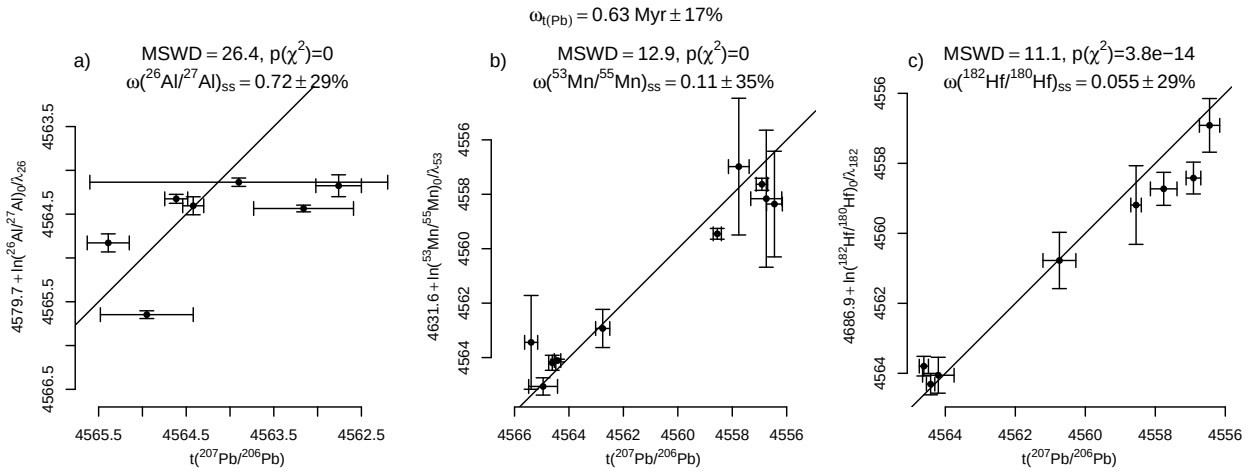


Figure 2: Pairwise comparison of the original Pb–Pb dates of Table 1 with Al–Mg (panel a), Mn–Cr (panel b) and Hf–W (panel c) dates for the same, rapidly cooled meteorites. The 1:1 lines were obtained using the optimal O-values (Equation 2). All three datasets are overdispersed around this line with respect to the analytical uncertainties (MSWD $\gg$ 1). ω-values are the estimated standard deviations of the excess dispersion, assuming normality (Section 4). This dispersion can either be attributed to the Pb–Pb data, or to the initial $(^{26}Al/^{27}Al)_{SS}$, $(^{53}Mn/^{55}Mn)_{SS}$ or $(^{182}Hf/^{180}Hf)_{SS}$-values, or a combination. Uncertainties are reported as standard errors.

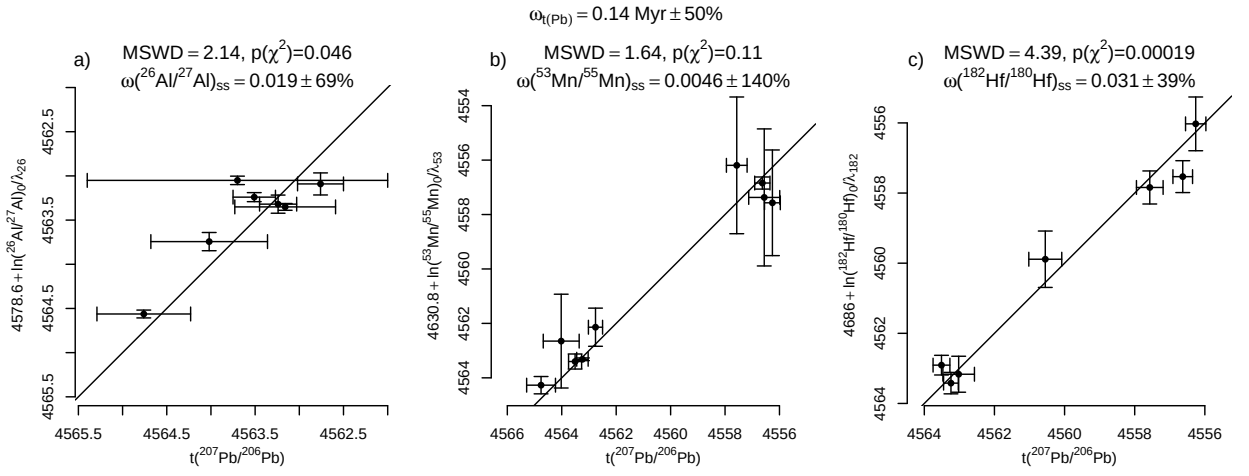


Figure 3: Same as Figure 2, but using the adjusted Pb–Pb dates of Table 1.

4 Quantifying dispersion

The observed overdispersion can be mathematically captured by random-effects models that attribute the excess dispersion to either the x- or the y-variables of Figures 2 and 3. The problem can be formalized by introducing the following definitions:

- $t_x(m,i)$: the 'measured' Pb–Pb age of the ith achondrite;
- $t_y(m,i)$: the 'measured' Al–Mg, Mn–Cr or Hf–W age of the ith achondrite;
- $\delta_x(i)$: the random offset between the measured x-value and the true Pb–Pb age, caused by late isotopic closure, or by analytical variations;
- $\delta_y(i)$: the random offset between the measured y-value and the true age, due to variations in isotopic closure time, or systematic analytical variations, or by heterogeneity of $(^{26}\text{Al}/^{27}\text{Al})_{\text{SS}}$, $(^{53}\text{Mn}/^{55}\text{Mn})_{\text{SS}}$, or $(^{182}\text{Hf}/^{180}\text{Hf})_{\text{SS}}$, respectively;
- $x(i)(+\delta_x(i))$: the true Pb–Pb age of the ith achondrite;
- $y(i)(+\delta_y(i))$: the true Al–Mg, Mn–Cr or Hf–W age of the ith achondrite;
- $\epsilon_x(i), \epsilon_y(i)$: residuals, caused by analytical uncertainty.

Using these definitions, we formulate two random-effects models:

$$\text{Model 3x: } \begin{cases} t_x(m,i) = x(i) + \delta_x(i) + \epsilon_x(i) \\ t_y(m,i) = x(i) + \epsilon_y(i) \end{cases} \text{, Model 3y: } \begin{cases} t_x(m,i) = y(i) + \epsilon_x(i) \\ t_y(m,i) = y(i) + \delta_y(i) + \epsilon_y(i) \end{cases} \quad (3)$$

To keep the problem tractable, it is necessary to make the simplifying assumption that $\delta_x(i)$, $\delta_y(i)$, $\epsilon_x(i)$ and $\epsilon_y(i)$ were drawn from normal distributions with zero mean. For $\epsilon_x(i)$ and $\epsilon_y(i)$, this assumption is reasonable. For $\delta_x(i)$ and $\delta_y(i)$, that is not necessarily so. For example, it is possible that there were multiple reservoirs with distinctly different $(^{26}\text{Al}/^{27}\text{Al})_{\text{SS}}$-values. However, Figure 3a) only contains 7 double-dates, which is too small to identify such multi-modality. Therefore, the simple normal assumption is necessary. The model can easily be modified to capture more realistic scenarios should a sufficiently large database of multi-dated meteorites become available in the future.

For model 3x, we will assume that the $\delta_x(i)$-values were drawn from a normal distribution with zero mean and standard deviation $\omega_{t(Pb)}$ (units: Myr). For model 3y, we will assume that the $\delta_y(i)$-values were drawn from a normal distribution with zero mean and standard deviation $\omega_y\lambda_y$ (units: none), which captures the dispersion of the $(^{26}\text{Al}/^{27}\text{Al})_{\text{SS}}$, $(^{53}\text{Mn}/^{55}\text{Mn})_{\text{SS}}$ or $(^{182}\text{Hf}/^{180}\text{Hf})_{\text{SS}}$-values, depending on the dataset used to solve the problem. Note that model 3x can be solved using the combined dataset of Al–Mg, Mn–Cr and Hf–W measurements.

If all of the overdispersion is attributed to heterogeneities in the abundances of short-lived radionuclides, the level of overdisperion is explained by:

1. 0.5–7.5% heterogeneity of the $(^{26}\text{Al}/^{27}\text{Al})_{\text{SS}}$-ratio (at 2σ, i.e. $\exp[\ln(0.019) \pm 2 \times 0.69]$; Figure 3a).
2. 0.0–7.6% heterogeneity of the $(^{53}\text{Mn}/^{55}\text{Mn})_{\text{SS}}$-ratio;
3. 1.4–6.8% heterogeneity of the $(^{182}\text{Hf}/^{180}\text{Hf})_{\text{SS}}$-ratio.

If one attributed the overdispersions in the Al–Mg, Mn–Cr, or Hf–W systems to analytical variations such as differences between groups in calibrations, then the same variations would apply. If the overdispersion is attributed to variations in the time of isotopic closure, of the Pb–Pb system in particular, then the data are explained by:

4. 50–380 ka excess dispersion in the Pb–Pb data.

The same excess dispersion in the Pb–Pb data would apply if the variations were attributed to analytical problems, such as improper correction for uranium isotopic fractionation, or over-zealous outlier rejection, which unduly promotes precision over accuracy (Vermeesch, 2026). We note that if the original Pb–Pb data are used, then the dispersion increases to 40–130% for $(^{26}\mathrm{Al}/^{27}\mathrm{Al})_{\mathrm{SS}}$, 5–22% for $(^{53}\mathrm{Mn}/^{55}\mathrm{Mn})_{\mathrm{SS}}$, 3–10% for $(^{182}\mathrm{Hf}/^{180}\mathrm{Hf})_{\mathrm{SS}}$ and 450–900 kyr for t_{Pb}. We also note that all the Hf–W data in Table 1 were obtained from angrites, which are thought to derive from a common parent body. Their overdispersion therefore cannot be attributed to nebular heterogeneity, suggesting instead that it reflects analytical effects. These effects could apply to the Hf–W data, the Pb–Pb data, or both.

5 Conclusion

In this paper we have presented two equivalent approaches to statistical chronometry, using an existing compilation of multi-dated meteorites originally presented by Desch et al. (2023b). The first approach tracks events forward in time from t_{SS}, which is treated as an unknown parameter, and is defined to be when $(^{26}\mathrm{Al}/^{27}\mathrm{Al})_{\mathrm{SS}} = 5.23 \times 10^{-5}$. Optimal values were found for $(^{53}\mathrm{Mn}/^{55}\mathrm{Mn})_{\mathrm{SS}}$ and $(^{182}\mathrm{Hf}/^{180}\mathrm{Hf})_{\mathrm{SS}}$ at $t = 0$; the Pb–Pb age of $t = 0$, t_{SS}; and the decay constant λ_{53}. The optimal values found by Desch et al. (2023b) are unaffected by the number of degrees of freedom, but the goodness of the fit is. After using the correct number of degrees of freedom, this approach reveals a small but statistically significant dispersion between the extinct and extant chronometers, even after adjustment for $^{238}\mathrm{U}/^{235}\mathrm{U}$ fractionation.

The second approach uses bivariate linear regression of doubly-dated meteorites to assess the concordance of the meteorite database. It does not treat t_{SS}, $(^{53}\mathrm{Mn}/^{55}\mathrm{Mn})_{\mathrm{SS}}$, and $(^{182}\mathrm{Hf}/^{180}\mathrm{Hf})_{\mathrm{SS}}$ as explict unknowns, but instead converts the $\ln\left(^{26}\mathrm{Al}/^{27}\mathrm{Al}\right)_0$, $\ln\left(^{53}\mathrm{Mn}/^{55}\mathrm{Mn}\right)_0$ and $\ln\left(^{182}\mathrm{Hf}/^{180}\mathrm{Hf}\right)_0$ logratios to absolute ages in 'Pb–Pb time' using a linear transformation (Equation 2). Applying the second approach to Desch et al. (2023a,b)'s database leads to the same conclusion as the first approach, that there statistically significant dispersion of the extinct and the extant chronometers. Adjustment of the Pb–Pb dates for $^{238}\mathrm{U}/^{235}\mathrm{U}$ fractionation reduces the overdispersion but does not completely remove it.

The regression approach marks a step forward in the development of a statistical chronometry for meteorites, by going from a simple test for statistical homogeneity, to the quantification of heterogeneity. This is important because, to paraphrase statistician George Box "all effects are significant (in some decimal place)". Given a sufficiently large or sufficiently precise database of multi-dated meteorites, even the smallest degree of heterogeneity eventually becomes detectable. The chi-square test cannot by itself determine the cause of the overdispersion, but is a useful tool. The overdispersion could be physical: the radionuclides could be heterogeneously distributed among achondrite parent bodies, or the isotopic systems may have closed at different times or been disturbed. Or the overdispersion could be analytical, due to discrepancies between groups in how isochrons are built.

Physical causes can be tested by examining the angrites separately, because they formed from the same parent body. Assuming that these meteorites remained undisturbed after their formation, their extinct and extant chronometers should be concordant. However, eight paired angrite Hf–W and Pb–Pb dates show that this is not the case. This suggests the presence analytical bias in the isotopic ratio data. Further investigation of these conclusions would benefit from improved analytical and statistical techniques. Analytical improvements include the addition of $^{238}\mathrm{U}/^{235}\mathrm{U}$ measurements in the same mineral phase as the Pb-isotope measurements. Although this is challenging to do in small or U-poor samples, it is highly beneficial as it removes the need for the nominal 0.3‰ correction applied by Desch et al. (2023a,b) to many of the meteorites in Table 1.

Statistical improvements include a more rigorous selection of aliquots to include in Pb–Pb isochrons. For example, Desch et al. (2023a) present a detailed discussion of Schiller et al. (2015)'s Pb–Pb dataset for NWA 1670 and show that there are at least three statistically equivalent ways to form an isochron through these data, with significantly different ages. To solve this problem, it would be useful to abandon the binary notion that aliquots fall either 'on' or 'off' the concordia line. A statistically more rigorous approach would be to assign a finite probability to both categories for each aliquot, similar to the random effects model of Section 4.

The future addition of more multi-dated meteorites to Desch et al. (2023b)'s compilation will make it possible to replace the simple random effects model of Section 4 with more realistic alternatives. For example, given a sufficient number of multi-dated NC and CC chondrites, it would be possible to quantify the within-group and between-group variability of these two meteorite classes. Thus, the nascent field of statistical chronometry has the potential to test physics-based predictions of solar-system formation and provide new insights into its temporal evolution.

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